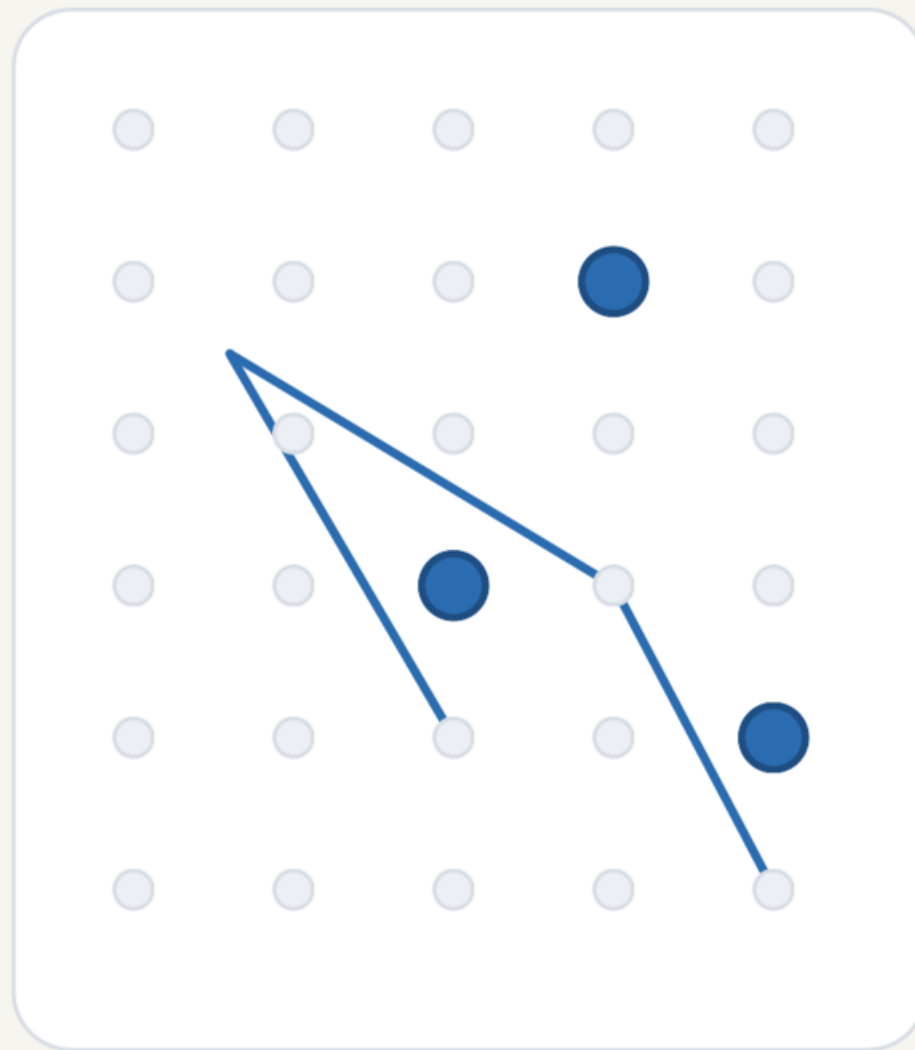


Sparse Separable Memory for Sequence Models

Can a sequence model gain large, adaptable memory without activating all of it for every token?

PROFESSOR DISCUSSION DECK

July 2026



Large memory · small active path

The capacity-computation tension

DENSE PATH

Every token touches the same large block.



Stored
capacity
 $\approx$
active
work

SPARSE MEMORY

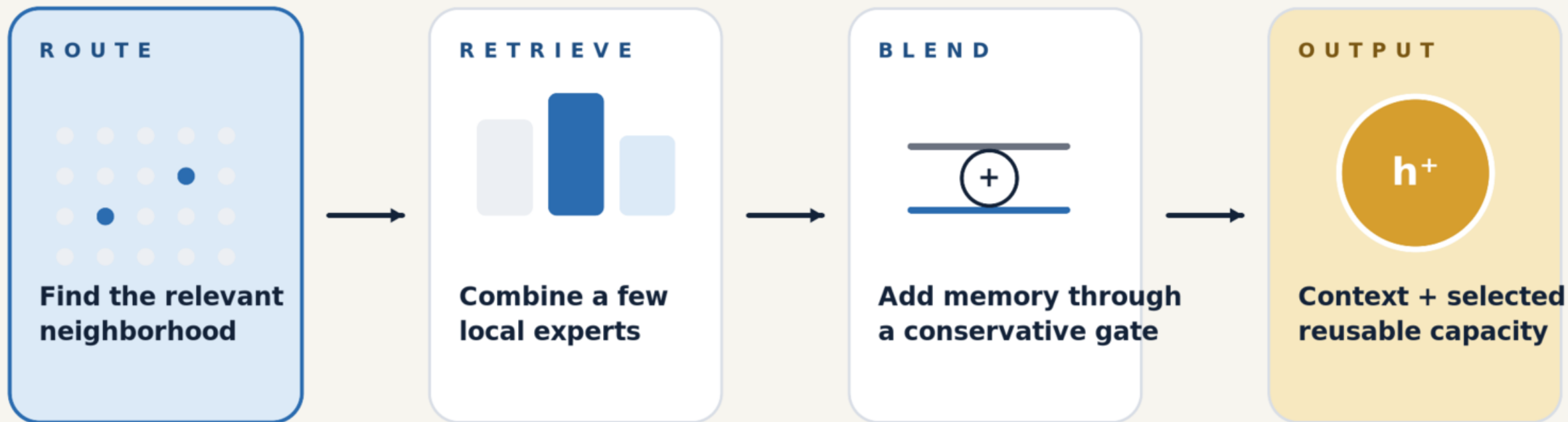
Every token consults only a few entries.



Large
potential
memory
+
small
active path

Decouple how much the model can store from how much it must activate.

The core intuition: route, retrieve, blend



Large potential memory · small active path · safe residual entry

One technical slide: sparse separable KAM

THREE LINES OF NOTATION

- 1 $I(z) = \text{TopK}(\text{similarity}(z, \text{keys}))$
- 2 $\text{memory}(u) = \sum \text{selected weight} \times \text{local expert}(u)$
- 3 $\text{output} = \text{Transformer path} + \text{small FFN} + \text{gate}$

WHY "SEPARABLE"?

GEOMETRY

Where should
the model look?



ALGEBRA

What should it
return there?

**Different objects
can learn on
different timescales.**

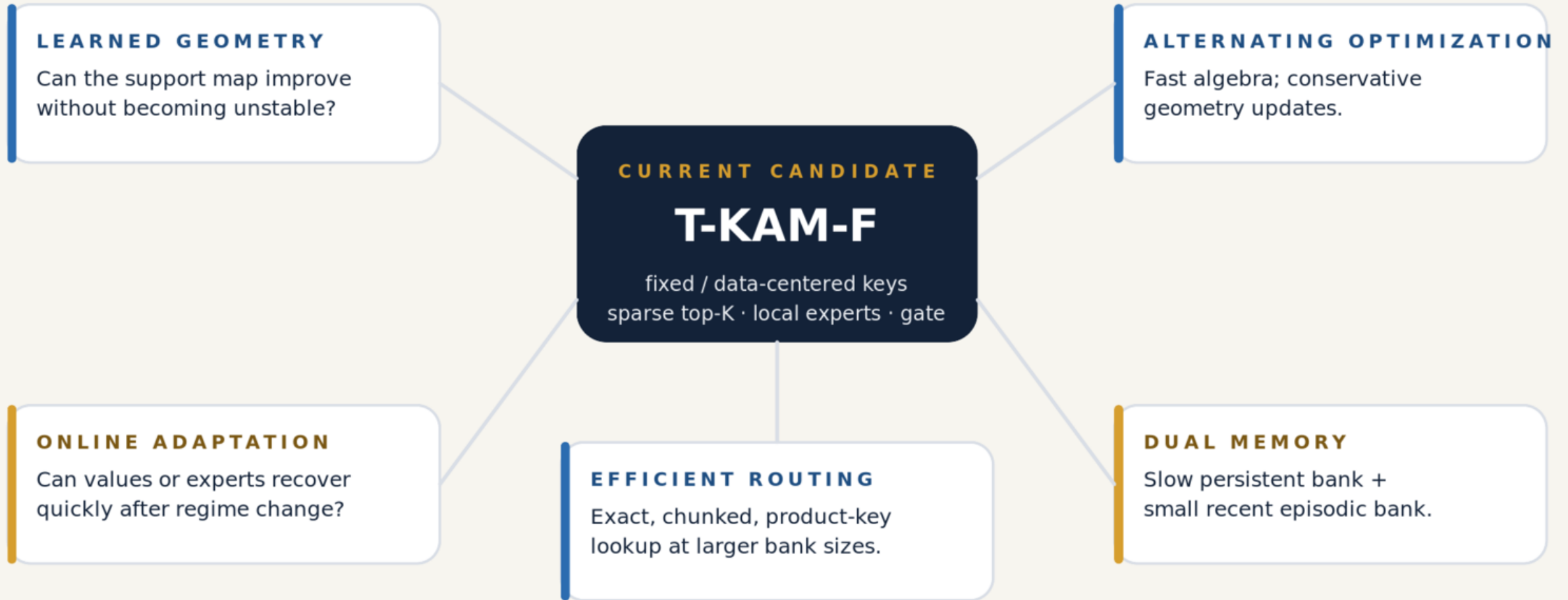
FAIRNESS Same data + token budget · matched total parameters · active parameters/FLOPs reported · strong controls

Two timescales: learn the map, then stabilize it

TRAINING PROGRESS



Current method and the research branches



Evidence is branch-specific: a fixed-key win cannot validate the other hypotheses.

What would count as convincing evidence?

156**fixed experiment rows**

four NVIDIA L4 GPUs

30**primary paired seeds**

TinyStories · KAM-F vs T-WIDE

24**replication pairs**

Tiny Shakespeare

50M**training tokens / row**

held-out endpoint

A prespecified evidence gate

FAIR

same data + budget

**HELD-OUT**

test endpoint

**PAIRED**

matched seeds

**REPLICATED**

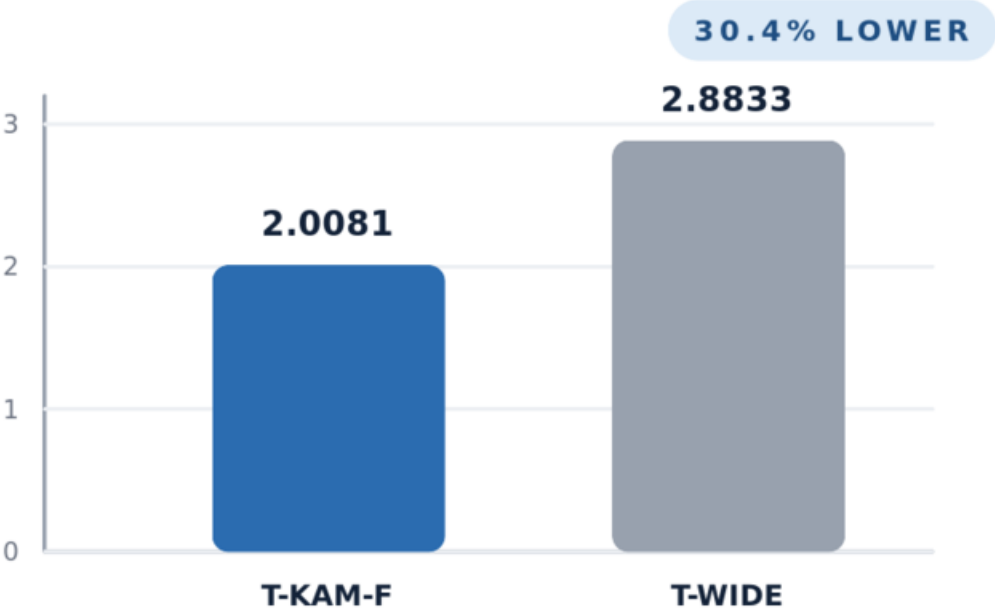
second corpus

2% MARGINconfidence interval must
clear the threshold**Primary + replication must both pass · no optional stopping · no seed replacement**

Intermediate evidence: promising direction, not a conclusion

Pilot validation loss at registered checkpoint

Lower is better · n = 3 paired seeds



Directional only

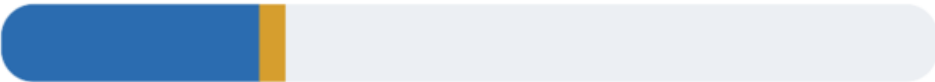
exact $p = 0.25$ · Holm-adjusted $p = 0.75$

Confirmation campaign status

July 28, 2026 · 6:40 PM EDT

43 / 156

complete



43 complete 4 running 109 pending

0 failures

Partial confirmation effects remain uninspected until the complete fixed sample finishes.

Promising enough to confirm; not strong enough to conclude.

The decision after confirmation

01

ROBUST FIXED-KEY ADVANTAGE

Accelerate T-KAM-F.
Scale memory banks and
test larger settings.

02

FIXED HELPS; LEARNED FAILS

Keep stable geometry.
Stop learned-memory
benefit claims.

03

NO REPLICATED QUALITY EDGE

Reject the general claim.
Retain only supported
niches or diagnostics.

Questions for discussion

- Is a 2% loss improvement the right scientific threshold?
- Should scale-up optimize quality, active compute, or adaptation?
- Which failure mode would be most informative for the broader community?

**The goal is a narrow,
testable claim—
not a universal memory architecture claim.**